Lesson 1.01 **Duration: 40 minutes**

What is ICT?

Learning Objectives

Engage (5-7 minutes)

Teacher Activity

Ask students: ‘Have you ever seen a computer? What did it do?’ Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: ‘All these involve processing information — that is what computers do.’

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: ‘What do you think this does?’ Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Memory units use powers of 10 (KB, MB, GB) - place value
Science: Electricity powers computer components - circuits
Language: Labeling diagrams builds technical vocabulary

Digital Citizen Reminder: Handle computer equipment carefully - it is expensive and shared by all students.

Resources Needed:
- Computer diagram or real computer
- labeled chart
- exit ticket paper

Differentiation

Approaching: Match pictures of computer parts to their names using a word bank
On Level: Label the computer diagram from memory and explain each part
Advanced: Create a buying guide for a school computer lab - list parts and prices
SEND: Touch and identify real computer parts with teacher guidance

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition - breaking down a computer into input, processing, memory, and output

Dormitory Task (Paper-and-Pencil)

Draw a labeled diagram of a computer from memory. Mark each part and write one sentence about what it does.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.02 **Duration: 40 minutes**

History of Computers

Learning Objectives

- Identify major generations of computers
- Explain how computers have changed over time
- Compare early computers with modern ones

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Timeline as a number line - understanding centuries and decades
Science: Evolution of technology mirrors biological evolution
Language: Writing historical narratives builds chronological writing skills

Digital Citizen Reminder: Appreciate how computing has evolved - modern computers fit in your pocket.

Resources Needed:
- Timeline chart
- picture cards of old computers
- worksheet

Differentiation

Approaching: Match pictures of old and new computers to their correct time period
On Level: Create a timeline of computer evolution with 5 milestones
Advanced: Research and compare processing speeds of 1950s vs modern computers
SEND: Sort computer pictures from oldest to newest with teacher guidance

Formative Assessment

Method: Timeline activity and oral presentation
CT Skill Assessed: Pattern Recognition - identifying trends in computer evolution

Dormitory Task (Paper-and-Pencil)

Draw a timeline showing at least 4 generations of computers. Label each with the year and one key feature.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.03 **Duration: 40 minutes**

Introduction to Internet

Learning Objectives

- Define the Internet and its basic function
- Explain how devices connect through networks
- Identify services available on the Internet

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit → CPU ALU + Control Unit → Memory → Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Network topology relates to graph theory in mathematics
Science: How data travels through cables and waves - physics of signals
Language: Writing about how the internet works builds expository writing

Digital Citizen Reminder: Never share personal information online - keep your full name, address, and school private.

Resources Needed:

- Internet diagram
- network map
- safety rules poster

Differentiation

Approaching: Match internet terms to their definitions using a word bank
On Level: Describe the journey of an email from sender to receiver
Advanced: Compare different internet connection types and their speeds
SEND: Identify parts of a network diagram with teacher guidance

Formative Assessment

Method: Diagram labeling and matching quiz
CT Skill Assessed: Decomposition - understanding how data moves from one computer to another

Dormitory Task (Paper-and-Pencil)

Draw a simple diagram showing how two computers connect through the internet. Label each part.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.04 **Duration: 40 minutes**

Web Browsers

Learning Objectives

- Identify popular web browsers and their icons
- Open a browser and navigate to a website
- Explain the parts of a browser window

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Understanding URL structure relates to pattern recognition
Science: Browsers render code into visual pages - computer science
Language: Writing clear instructions for browser use - procedural text

Digital Citizen Reminder: Keep your browser updated - updates fix security problems and add new features.

Resources Needed:
- Browser screenshot
- toolbar labeled diagram
- practice worksheet

Differentiation

Approaching: Identify browser toolbar buttons from a labeled diagram
On Level: Navigate to three different websites using the address bar
Advanced: Compare two different browsers and create a feature comparison table
SEND: Use browser features with teacher guidance - back, forward, refresh

Formative Assessment

Method: Practical navigation task
CT Skill Assessed: Algorithm Design - following steps to complete browser tasks

Dormitory Task (Paper-and-Pencil)

Draw a web browser window and label the address bar, back button, forward button, and refresh button.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.05 **Duration: 40 minutes**

Search Engines

Learning Objectives

- Define search engines and name popular ones
- Perform a basic search with keywords
- Evaluate search results for relevance

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Search results ranked by relevance - understanding algorithms
Science: How search engines crawl and index the web - information science
Language: Writing effective search queries builds precise language skills

Digital Citizen Reminder: Not all search results are reliable - check if information comes from a trusted source.

Resources Needed:
- Computer with internet
- search tips handout
- worksheet

Differentiation

Approaching: Type simple search queries and click on results
On Level: Use three search techniques (quotes, minus, site:) to find specific information
Advanced: Evaluate search results for reliability and compare different sources
SEND: Perform a search with teacher guidance using step-by-step instructions

Formative Assessment

Method: Search skills practical test
CT Skill Assessed: Evaluation - assessing the quality and reliability of search results

Dormitory Task (Paper-and-Pencil)

Write five different search queries you would use to find information about your school. Explain which one would give the best results.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.06 **Duration: 40 minutes**

URLs and Websites

Learning Objectives

- Define URL and identify its parts
- Distinguish between websites and web pages
- Navigate using URLs in a browser

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: URL structure follows specific rules - understanding protocols and syntax
Science: Websites are hosted on servers located worldwide - geography
Language: Writing clear web content - audience awareness and structure

Digital Citizen Reminder: Check URLs carefully before clicking - fake websites can steal your information.

Resources Needed:
- URL anatomy diagram
- website examples
- labeling worksheet

Differentiation

Approaching: Identify the parts of a URL from a labeled diagram
On Level: Classify websites as educational, entertainment, or commercial
Advanced: Compare URLs to identify secure (https) vs non-secure connections
SEND: Read and identify URL components with teacher guidance

Formative Assessment

Method: URL labeling quiz
CT Skill Assessed: Abstraction - understanding that URLs follow a structured format

Dormitory Task (Paper-and-Pencil)

Write the URL of your favorite website. Label the protocol, domain name, and page name.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.07 **Duration: 40 minutes**

Introduction to GPS

Learning Objectives

- Define GPS and explain how it works
- Identify devices that use GPS technology
- Describe real-life applications of GPS

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit → CPU ALU + Control Unit → Memory → Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: GPS uses latitude and longitude coordinates - geography and math
Science: Satellites orbit Earth using principles of physics and gravity
Language: Writing directions builds spatial language and sequential writing

Digital Citizen Reminder: GPS tracks your location - be aware of privacy settings on your devices.

Resources Needed:
- GPS device or phone with GPS
- map
- coordinate worksheet

Differentiation

Approaching: Read latitude and longitude from a simple map
On Level: Use GPS coordinates to locate three places on a map
Advanced: Explain how GPS satellites determine your exact position
SEND: Find your location on a map with teacher guidance

Formative Assessment

Method: Coordinate reading activity
CT Skill Assessed: Decomposition - breaking down how GPS determines location using satellites

Dormitory Task (Paper-and-Pencil)

Write the latitude and longitude of your school. Explain in your own words how GPS finds this location.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.08 **Duration: 40 minutes**

Google Earth

Learning Objectives

- Open Google Earth and navigate the interface
- Search for locations and explore them virtually
- Measure distances between two points

Engage (5-7 minutes)

Teacher Activity

Ask students: ‘Have you ever seen a computer? What did it do?’ Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: ‘All these involve processing information — that is what computers do.’

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: ‘What do you think this does?’ Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: ‘An electronic device that accepts data, processes it, and produces output.’ Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Understanding scale and zoom levels - mathematical ratios
Science: Satellite imagery and remote sensing - earth science
Language: Writing travel guides requires descriptive and persuasive writing

Digital Citizen Reminder: Respect privacy - do not zoom into private properties or share others' locations.

Resources Needed:
- Computer with Google Earth
- location coordinates
- exploration worksheet

Differentiation

Approaching: Navigate to your school using the search bar
On Level: Measure the distance between two cities using the ruler tool
Advanced: Create a virtual tour of five landmarks with descriptions
SEND: Explore locations with teacher guidance using basic navigation

Formative Assessment

Method: Virtual tour presentation
CT Skill Assessed: Evaluation - assessing the accuracy and usefulness of satellite imagery

Dormitory Task (Paper-and-Pencil)

Use Google Earth to find your school. Write the name of three nearby landmarks and estimate their distance from your school.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.09 **Duration: 40 minutes**

Internet Safety

Learning Objectives

- Identify common online dangers
- Practice safe browsing habits
- Create a strong password

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Risk assessment in math - probability and consequences
Science: Understanding online threats relates to biology (viruses as metaphor)
Language: Writing safety rules builds persuasive and instructional writing

Digital Citizen Reminder: Never share your password with anyone except your parents or teacher. Use strong passwords with letters, numbers, and symbols.

Resources Needed:
- Safety rules poster
- scenario cards
- pledge template

Differentiation

Approaching: Match online safety rules to their correct category
On Level: Identify safe and unsafe behaviors in five online scenarios
Advanced: Create a personal internet safety plan with at least 8 rules
SEND: Discuss safety rules with teacher guidance using picture prompts

Formative Assessment

Method: Safety pledge and scenario quiz
CT Skill Assessed: Evaluation - assessing online situations for safety risks

Dormitory Task (Paper-and-Pencil)

Write ten internet safety rules. For each rule, explain what could happen if you do not follow it.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.10 **Duration: 40 minutes**

Email Basics

Learning Objectives

- Identify the parts of an email address
- Compose and send a simple email
- Explain the purpose of inbox, sent, and draft folders

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit ⟶ CPU ALU + Control Unit ⟶ Memory ⟶ Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Understanding email structure - formal letter writing in language
Science: Email uses addresses like mathematical notation
Language: Writing clear subject lines and messages builds concise writing

Digital Citizen Reminder: Do not open emails from strangers - they may contain viruses or scams.

Resources Needed:
- Email template
- sample emails
- practice worksheet

Differentiation

Approaching: Read a sample email and identify the sender, subject, and body
On Level: Compose a simple email with subject line and greeting
Advanced: Write a formal email with proper structure, tone, and formatting
SEND: Send an email with teacher guidance - filling in To, Subject, and Body

Formative Assessment

Method: Email composition task
CT Skill Assessed: Algorithm Design - following the correct steps to compose and send an email

Dormitory Task (Paper-and-Pencil)

Write an email to your teacher asking for permission to use the computer lab. Include To, Subject, and body.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.11 **Duration: 40 minutes**

Downloading Files

Learning Objectives

- Download files from the Internet safely
- Identify file types and their extensions
- Organize downloaded files in folders

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit → CPU ALU + Control Unit → Memory → Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: ‘Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?’ Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: ‘1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.’ Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: File sizes in bytes, KB, MB - unit conversion and estimation
Science: Understanding how files travel through networks - physics
Language: Following download instructions builds procedural writing skills

Digital Citizen Reminder: Only download files from trusted websites. Scan all downloads for viruses before opening.

Resources Needed:
- Computer with internet
- download folder
- practice files

Differentiation

Approaching: Download a file and save it to the correct folder
On Level: Download three files and organize them by type
Advanced: Manage download settings and file storage efficiently
SEND: Download and save a file with teacher guidance step by step

Formative Assessment

Method: Practical download and organize task
CT Skill Assessed: Decomposition - breaking down the download process into steps

Dormitory Task (Paper-and-Pencil)

Write the steps to download a file from the internet. List at least 5 steps in order.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.12 **Duration: 40 minutes**

Chapter 1 Review

Learning Objectives

- Review key concepts from Chapter 1
- Demonstrate skills learned in previous lessons
- Identify areas needing more practice

Engage (5-7 minutes)

Teacher Activity

Ask students: 'Have you ever seen a computer? What did it do?' Write all responses on the board. Group them into categories: typing, playing games, watching videos, calculating. Guide discussion: 'All these involve processing information — that is what computers do.'

Student Activity

Raise hands and share experiences. Observe the pattern on the board. Suggest additional uses: printing, internet browsing.

Explore (10-12 minutes)

Teacher Activity

Show the parts of a computer on a diagram or real machine. Point to each part and ask: 'What do you think this does?' Let students guess before explaining. Identify: monitor output, keyboard input, mouse input, CPU processing, speaker output.

Student Activity

Observe the diagram. Write down the name and function of each part. Discuss with a partner which parts are input and which are output.

Explain (10-12 minutes)

Teacher Activity

Define computer: 'An electronic device that accepts data, processes it, and produces output.' Draw the block diagram: Input Unit → CPU ALU + Control Unit → Memory → Output Unit. Explain each unit with village examples: farmer enters data input, computer calculates processing, result displays output.

Student Activity

Take notes. Draw the block diagram. Label each unit. Give examples: ATM input: card, processing: verify, output: cash.

Elaborate (8-10 minutes)

Teacher Activity

Present scenario: 'Your school needs to maintain records for 200 students — marks, attendance, fees. How would you do this without a computer? How would you do it with a computer?' Compare both approaches.

Student Activity

Work in pairs. List steps for manual method register, pen, manual counting vs computer method enter data, auto-calculate, print. Discuss which is faster and more accurate.

Evaluate (5-8 minutes)

Teacher Activity

Dictate exit ticket questions: '1. Name three parts of a computer. 2. What does CPU stand for? 3. Give one example of input and one of output.' Collect tickets as students leave.

Student Activity

Complete the three questions individually. Review answers before submitting.

Cross-Curricular Connections

Mathematics: Review exercises reinforce mathematical problem-solving
Science: Chapter review connects all internet and computing concepts
Language: Writing summaries builds synthesis and review skills

Digital Citizen Reminder: Apply internet safety rules you learned in every online activity.

Resources Needed:
- Review worksheet
- answer key
- chapter summary

Differentiation

Approaching: Complete fill-in-the-blank review questions
On Level: Answer short answer questions about chapter concepts
Advanced: Create a concept map connecting all chapter topics
SEND: Review key terms with teacher guidance using flashcards

Formative Assessment

Method: Chapter review quiz
CT Skill Assessed: Evaluation - self-assessing understanding of chapter concepts

Dormitory Task (Paper-and-Pencil)

Write a summary of everything you learned in this chapter. Include at least 8 key points.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.01	Duration: 40 minutes

Introduction to Graphics

Learning Objectives

- Define graphics and identify their types
- Distinguish between raster and vector graphics
- Name software used for creating graphics

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Pixel art uses grid coordinates - connecting to coordinate geometry
Science: Color mixing in digital art relates to light science
Language: Describing visual elements builds art vocabulary

Digital Citizen Reminder: Respect copyright - do not copy others artwork without permission.

Resources Needed:
- Computer with Paint
- color wheel
- sample graphics

Differentiation

Approaching: Identify basic shapes and colors in digital images
On Level: Create a simple drawing using at least 5 different tools
Advanced: Design a complex graphic combining multiple techniques
SEND: Use basic drawing tools with teacher guidance

Formative Assessment

Method: Tool identification and basic drawing task
CT Skill Assessed: Decomposition - breaking complex images into simple shapes

Dormitory Task (Paper-and-Pencil)

Draw a picture of your home using only basic shapes - circles, squares, triangles. Label each shape.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.02 **Duration: 40 minutes**

Paint Software Basics

Learning Objectives

- Open Paint and identify its interface components
- Use basic tools like pencil, brush, and eraser
- Save a drawing in Paint

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Understanding tool functions relates to following mathematical procedures
Science: Digital brushes simulate real art materials - materials science
Language: Writing tool guides builds instructional writing

Digital Citizen Reminder: Save your artwork frequently to avoid losing your work.

Resources Needed:
- Computer with Paint
- toolbar diagram
- practice template

Differentiation

Approaching: Use the pencil tool to draw a simple house
On Level: Create a picture using at least 4 different tools
Advanced: Design a landscape using multiple tools and techniques
SEND: Draw with different tools and identify their effects with teacher guidance

Formative Assessment

Method: Tool usage demonstration
CT Skill Assessed: Algorithm Design - following steps to create artwork

Dormitory Task (Paper-and-Pencil)

Write the names of five Paint tools and what each one does.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.03 **Duration: 40 minutes**

Drawing Tools

Learning Objectives

- Use shape tools to create geometric figures
- Apply line tools with different thicknesses
- Combine shapes to create simple compositions

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Geometric shapes in art - identifying mathematical shapes in drawings
Science: Understanding brush strokes relates to fluid dynamics
Language: Describing drawing techniques builds technical vocabulary

Digital Citizen Reminder: Use drawing tools responsibly - do not draw on screens with markers or sharp objects.

Resources Needed:
- Computer with Paint
- tool practice sheets
- example artwork

Differentiation

Approaching: Use the line and rectangle tools to draw a building
On Level: Create a picture using at least 6 different drawing tools
Advanced: Combine advanced tools to create a detailed scene
SEND: Practice drawing with each tool and describe the results

Formative Assessment

Method: Drawing portfolio and tool checklist
CT Skill Assessed: Pattern Recognition - using repeated elements in designs

Dormitory Task (Paper-and-Pencil)

Draw a picture using only straight lines and rectangles. Count and write how many of each you used.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.04 **Duration: 40 minutes**

Color Selection

Learning Objectives

- Identify primary and secondary colors
- Use the color palette and color picker
- Apply colors to shapes and backgrounds

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Color theory - primary and secondary colors, mixing ratios
Science: How screens produce color using RGB light - physics
Language: Describing colors builds descriptive and sensory vocabulary

Digital Citizen Reminder: Choose appropriate colors for your artwork - some color combinations are hard to read.

Resources Needed:
- Color wheel
- color palette samples
- practice artwork

Differentiation

Approaching: Fill shapes with correct colors from a guide
On Level: Create a color palette and use it in a drawing
Advanced: Apply color theory to create mood and emphasis in artwork
SEND: Select and apply colors with teacher guidance

Formative Assessment

Method: Color matching activity
CT Skill Assessed: Abstraction - understanding that digital colors combine red, green, and blue

Dormitory Task (Paper-and-Pencil)

Draw a color wheel showing primary and secondary colors. Label each color.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.05 **Duration: 40 minutes**

Symmetric Designs

Learning Objectives

- Create mirror images using the select tool
- Design patterns with rotational symmetry
- Apply symmetry to real-world design problems

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Symmetry in math - reflection, rotation, and translation
Science: Symmetry in nature - biology and natural forms
Language: Writing about symmetry requires precise geometric language

Digital Citizen Reminder: Symmetry tools create mirror images - think before you use them.

Resources Needed:
- Computer with Paint
- symmetry examples
- practice grid

Differentiation

Approaching: Create a simple mirror image using the select and copy tools
On Level: Design a symmetric pattern with 4-fold rotational symmetry
Advanced: Create a complex mandala combining reflection and rotation
SEND: Make a symmetric design with teacher guidance step by step

Formative Assessment

Method: Symmetry design assessment
CT Skill Assessed: Pattern Recognition - identifying and creating symmetrical patterns

Dormitory Task (Paper-and-Pencil)

Fold a paper in half and draw half a butterfly. Open it to see the full symmetric design.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.06 **Duration: 40 minutes**

Image Editing

Learning Objectives

- Crop images to remove unwanted areas
- Resize images while maintaining proportions
- Adjust brightness and contrast of images

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Crop dimensions relate to area and perimeter calculations
Science: Image compression reduces file size - understanding ratios
Language: Writing editing instructions builds precise procedural text

Digital Citizen Reminder: Only edit images you have permission to use - respect copyright.

Resources Needed:
- Computer with image editor
- sample images
- editing checklist

Differentiation

Approaching: Crop an image to remove unwanted parts
On Level: Resize, crop, and adjust brightness of an image
Advanced: Apply multiple edits including crop, resize, rotate, and color adjust
SEND: Edit an image with teacher guidance using basic tools

Formative Assessment

Method: Image editing task
CT Skill Assessed: Algorithm Design - following a sequence of editing steps

Dormitory Task (Paper-and-Pencil)

Write a list of 5 image editing operations and explain what each one does.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.07 **Duration: 40 minutes**

Layers Concept

Learning Objectives

- Understand what layers are in graphics software
- Create and manage multiple layers
- Arrange layer order to control visibility

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Layers stack like fractions - understanding ordering and precedence
Science: How transparent layers work - optics and light
Language: Explaining layers requires clear organizational writing

Digital Citizen Reminder: Understanding layers helps organize work - a skill useful beyond computing.

Resources Needed:

- Computer with layer-capable editor
- layer examples
- practice files

Differentiation

Approaching: Create two layers and draw different elements on each
On Level: Use layers to create a composite image with foreground and background
Advanced: Manage complex multi-layer compositions with blending modes
SEND: Create and manage layers with teacher guidance

Formative Assessment

Method: Layer management demonstration
CT Skill Assessed: Abstraction - understanding that layers separate elements for independent editing

Dormitory Task (Paper-and-Pencil)

Draw three separate elements (sun, house, tree) on separate pieces of paper. Stack them to create one picture.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.08 **Duration: 40 minutes**

Filters and Effects

Learning Objectives

- Apply basic filters like blur and sharpen
- Use artistic effects to transform images
- Compare original and filtered versions

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Filters transform images mathematically - understanding transformations
Science: Digital filters mimic photographic techniques - optics
Language: Describing visual effects builds descriptive vocabulary

Digital Citizen Reminder: Effects should enhance your work, not hide poor quality - aim for quality over effects.

Resources Needed:
- Computer with image editor
- filter gallery
- before-after examples

Differentiation

Approaching: Apply one filter and compare before and after
On Level: Apply three different effects and describe what each does
Advanced: Combine multiple effects to create an artistic style
SEND: Apply and compare filters with teacher guidance

Formative Assessment

Method: Effect application and comparison
CT Skill Assessed: Evaluation - assessing which effects improve an image

Dormitory Task (Paper-and-Pencil)

Write the names of four image effects and describe what each one does to a picture.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.09 **Duration: 40 minutes**

Text in Graphics

Learning Objectives

- Add text to graphics using text tools
- Format text with fonts, sizes, and colors
- Create simple banners and posters with text

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Text in Graphics - geometric transformations
Science: Scientific applications of Text in Graphics - digital imaging
Language: Writing about Text in Graphics builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Text in Graphics concepts through practical exercises
Advanced: Apply Text in Graphics concepts to solve complex problems independently
SEND: Practice Text in Graphics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Text in Graphics activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Text in Graphics into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple design using only basic shapes. Label the tools you would use to create it digitally.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.10 **Duration: 40 minutes**

File Formats

Learning Objectives

- Identify common image file formats
- Compare qualities and uses of different formats
- Save files in appropriate formats for specific purposes

Engage (5-7 minutes)

Teacher Activity

Ask: ‘How many of you have used the internet? What did you do?’ Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: ‘Information and Communication Technology — all tools that handle information.’

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: ‘What about at the bus stand? Hospital? Shop?’ Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Mathematical concepts in File Formats - geometric transformations
Science: Scientific applications of File Formats - digital imaging
Language: Writing about File Formats builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Advanced Graphics reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of File Formats concepts through practical exercises
Advanced: Apply File Formats concepts to solve complex problems independently
SEND: Practice File Formats skills with step-by-step teacher guidance

Formative Assessment

Method: Practical File Formats activity and oral questioning
CT Skill Assessed: Decomposition - breaking down File Formats into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about File Formats. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.11 **Duration: 40 minutes**

Project - School Logo

Learning Objectives

- Apply all learned graphics skills to create a logo
- Plan and execute a design project step by step
- Present and explain design choices to others

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: 'Which ICT tools help in each?' Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?'

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - School Logo - geometric transformations
Science: Scientific applications of Project - School Logo - digital imaging
Language: Writing about Project - School Logo builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply vector graphics and image editing skills responsibly and ethically.

Resources Needed:
- Advanced Graphics reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with vector graphics and image editing support
On Level: Demonstrate understanding of Project - School Logo concepts through practical exercises
Advanced: Apply Project - School Logo concepts to solve complex problems independently
SEND: Practice Project - School Logo skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - School Logo project

Dormitory Task (Paper-and-Pencil)

Plan a Project - School Logo project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 2.12 **Duration: 40 minutes**

Chapter 2 Review

Learning Objectives

- Review all graphics concepts from Chapter 2
- Demonstrate integrated skills in a final project
- Reflect on learning and identify growth areas

Engage (5-7 minutes)

Teacher Activity

Ask: 'How many of you have used the internet? What did you do?' Write responses. Group: searching, watching videos, chatting, email. Introduce ICT: 'Information and Communication Technology — all tools that handle information.'

Student Activity

Share internet experiences. Identify the pattern — all involve information. Suggest more examples: online classes, weather apps.

Explore (10-12 minutes)

Teacher Activity

Organize pairs. Give 3 minutes to list every ICT device and internet service they can think of. Prompt: 'What about at the bus stand? Hospital? Shop?' Count total unique items.

Student Activity

Brainstorm in pairs. Write: phones, TVs, radios, ATMs, email, WhatsApp, Google. Share with class.

Explain (10-12 minutes)

Teacher Activity

Define ICT components: Information data types, Communication sharing methods, Technology devices and software. Draw Venn diagram. Explain internet as a network connecting millions of computers worldwide.

Student Activity

Take notes. Copy the Venn diagram. Give personal examples for each component.

Elaborate (8-10 minutes)

Teacher Activity

Present 5 village scenarios: hospital records, bus timings, farmer weather, teacher-parent messages, shopkeeper inventory. Ask: ‘Which ICT tools help in each?’ Monitor group discussions.

Student Activity

Group work: match ICT tools to scenarios. Present one match to the class with reasoning.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. What does ICT stand for? 2. Name three uses of the internet. 3. How does ICT help your village?’

Student Activity

Write answers individually. Submit before leaving.

Cross-Curricular Connections

Mathematics: Review exercises consolidate programming concepts
Science: Chapter summaries connect related topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 2 Review skills responsibly and ethically.

Resources Needed:
- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 2 Review support
On Level: Demonstrate Chapter 2 Review through practical exercises
Advanced: Apply Chapter 2 Review to solve complex problems independently
SEND: Practice Chapter 2 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 2 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 2 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 2 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.01 **Duration: 40 minutes**

Types of Communication

Learning Objectives

- Define communication and its purpose
- Identify verbal, non-verbal, and digital communication
- Compare traditional and modern communication methods

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Types of Communication - network topology
Science: Scientific applications of Types of Communication - data transmission
Language: Writing about Types of Communication builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Networking reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Types of Communication concepts through practical exercises
Advanced: Apply Types of Communication concepts to solve complex problems independently
SEND: Practice Types of Communication skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Types of Communication activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Types of Communication into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Types of Communication. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.02 **Duration: 40 minutes**

Audio Communication

Learning Objectives

- Understand how sound is used for communication
- Identify devices used for audio communication
- Explain the difference between analog and digital audio

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Audio Communication - network topology
Science: Scientific applications of Audio Communication - data transmission
Language: Writing about Audio Communication builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Audio Communication concepts through practical exercises
Advanced: Apply Audio Communication concepts to solve complex problems independently
SEND: Practice Audio Communication skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Audio Communication activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Audio Communication into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.03 **Duration: 40 minutes**

Video Communication

Learning Objectives

- Understand how video combines audio and visual elements
- Identify devices and platforms for video communication
- Explain the basic components of a video file

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Video Communication - network topology
Science: Scientific applications of Video Communication - data transmission
Language: Writing about Video Communication builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Video Communication concepts through practical exercises
Advanced: Apply Video Communication concepts to solve complex problems independently
SEND: Practice Video Communication skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Video Communication activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Video Communication into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.04 **Duration: 40 minutes**

Introduction to Audacity

Learning Objectives

- Identify Audacity as a free audio editing tool
- Navigate the Audacity interface and key controls
- Record and play back a short audio clip

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Introduction to Audacity - network topology
Science: Scientific applications of Introduction to Audacity - data transmission
Language: Writing about Introduction to Audacity builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Introduction to Audacity concepts through practical exercises
Advanced: Apply Introduction to Audacity concepts to solve complex problems independently
SEND: Practice Introduction to Audacity skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Introduction to Audacity activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Introduction to Audacity into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Introduction to Audacity. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.05 **Duration: 40 minutes**

Recording Audio

Learning Objectives

- Set up microphone and adjust recording levels
- Record clear audio with proper techniques
- Identify and reduce common recording errors

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: ‘1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.’

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Recording Audio - network topology
Science: Scientific applications of Recording Audio - data transmission
Language: Writing about Recording Audio builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Recording Audio concepts through practical exercises
Advanced: Apply Recording Audio concepts to solve complex problems independently
SEND: Practice Recording Audio skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Recording Audio activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Recording Audio into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.06 **Duration: 40 minutes**

Editing Audio

Learning Objectives

- Cut, copy, and paste audio segments in Audacity
- Remove unwanted sections and silence from recordings
- Apply basic trimming to clean up audio

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Editing Audio - network topology
Science: Scientific applications of Editing Audio - data transmission
Language: Writing about Editing Audio builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Editing Audio concepts through practical exercises
Advanced: Apply Editing Audio concepts to solve complex problems independently
SEND: Practice Editing Audio skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Editing Audio activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Editing Audio into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.07 **Duration: 40 minutes**

Effects and Filters

Learning Objectives

- Apply audio effects like echo, reverb, and amplification
- Use noise reduction to improve audio quality
- Understand when and how to use different filters

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Effects and Filters - network topology
Science: Scientific applications of Effects and Filters - data transmission
Language: Writing about Effects and Filters builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Effects and Filters concepts through practical exercises
Advanced: Apply Effects and Filters concepts to solve complex problems independently
SEND: Practice Effects and Filters skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Effects and Filters activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Effects and Filters into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Effects and Filters. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.08 **Duration: 40 minutes**

Mixing Audio

Learning Objectives

- Combine multiple audio tracks into one
- Adjust volume levels and balance between tracks
- Use the timeline to align audio elements

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Mixing Audio - network topology
Science: Scientific applications of Mixing Audio - data transmission
Language: Writing about Mixing Audio builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Mixing Audio concepts through practical exercises
Advanced: Apply Mixing Audio concepts to solve complex problems independently
SEND: Practice Mixing Audio skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Mixing Audio activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Mixing Audio into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.09 **Duration: 40 minutes**

Exporting Audio

Learning Objectives

- Export edited audio in common formats like MP3 and WAV
- Choose appropriate quality settings for different uses
- Understand file size versus quality trade-offs

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Exporting Audio - network topology
Science: Scientific applications of Exporting Audio - data transmission
Language: Writing about Exporting Audio builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Exporting Audio concepts through practical exercises
Advanced: Apply Exporting Audio concepts to solve complex problems independently
SEND: Practice Exporting Audio skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Exporting Audio activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Exporting Audio into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.10 **Duration: 40 minutes**

Video Editing Basics

Learning Objectives

- Import video clips into an editing application
- Trim and arrange clips on a timeline
- Add transitions and text overlays to video

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Video Editing Basics - network topology
Science: Scientific applications of Video Editing Basics - data transmission
Language: Writing about Video Editing Basics builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Video Editing Basics concepts through practical exercises
Advanced: Apply Video Editing Basics concepts to solve complex problems independently
SEND: Practice Video Editing Basics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Video Editing Basics activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Video Editing Basics into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.11 **Duration: 40 minutes**

Project - Audio Story

Learning Objectives

- Plan and script a short audio story
- Record, edit, and mix voice narration with sound effects
- Export and present the final audio story

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Audio Story - network topology
Science: Scientific applications of Project - Audio Story - data transmission
Language: Writing about Project - Audio Story builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply network configuration skills responsibly and ethically.

Resources Needed:

- Networking reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with network configuration support
On Level: Demonstrate understanding of Project - Audio Story concepts through practical exercises
Advanced: Apply Project - Audio Story concepts to solve complex problems independently
SEND: Practice Project - Audio Story skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Audio Story project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Audio Story project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 3.12 **Duration: 40 minutes**

Chapter 3 Review

Learning Objectives

- Review all concepts from Audio \\\& Visual Communication
- Demonstrate audio recording, editing, and mixing skills
- Apply chapter knowledge to a practical scenario

Engage (5-7 minutes)

Teacher Activity

Show two images: one hand-drawn, one digital. Ask: 'Which looks better? Why?' Discuss how computers make art easier. Show MS Paint interface on projector.

Student Activity

Compare images. Identify differences: colors, neatness, erasing. Express curiosity about digital art tools.

Explore (10-12 minutes)

Teacher Activity

Open MS Paint. Demonstrate tools: pencil, brush, fill, text, shapes, eraser. Let students experiment on their own or paper simulation.

Student Activity

Explore Paint tools. Draw a simple picture using at least 3 different tools. Experiment with colors and shapes.

Explain (10-12 minutes)

Teacher Activity

Define graphics: visual representations of information. Types: painting, drawing, photography. Digital advantages: undo, layers, effects, easy sharing. Show file formats: BMP, JPG, PNG.

Student Activity

Take notes. Compare digital vs manual art advantages. List file formats and their uses.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Create a poster for your school using Paint. Include text, shapes, and at least 3 colors.' Show examples of good digital posters.

Student Activity

Work on poster. Apply tools learned. Get peer feedback on design.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two graphics software. 2. What is the advantage of digital art over hand-drawing? 3. Name one file format for images.'

Student Activity

Answer three questions. Review before submitting.

Cross-Curricular Connections

Mathematics: Review exercises consolidate database concepts
Science: Chapter summaries connect SQL topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 3 Review skills responsibly and ethically.

Resources Needed:
- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 3 Review support
On Level: Demonstrate Chapter 3 Review through practical exercises
Advanced: Apply Chapter 3 Review to solve complex problems independently
SEND: Practice Chapter 3 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 3 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 3 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 3 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.01 **Duration: 40 minutes**

Introduction to Data

Learning Objectives

- Define data and distinguish between data and information
- Identify different types of data in daily life
- Explain why computers represent data as numbers

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Introduction to Data - mathematical concepts
Science: Scientific applications of Introduction to Data - scientific applications
Language: Writing about Introduction to Data builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Introduction to Data concepts through practical exercises
Advanced: Apply Introduction to Data concepts to solve complex problems independently
SEND: Practice Introduction to Data skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Introduction to Data activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Introduction to Data into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Introduction to Data. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.02 **Duration: 40 minutes**

Number Systems

Learning Objectives

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: ‘How was this recorded? How do we share sound over distance?’ Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: ‘Record a 30-second introduction about yourself. Practice speaking clearly.’ If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Number Systems - mathematical concepts
Science: Scientific applications of Number Systems - scientific applications
Language: Writing about Number Systems builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Number Systems concepts through practical exercises
Advanced: Apply Number Systems concepts to solve complex problems independently
SEND: Practice Number Systems skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Number Systems activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Number Systems into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Number Systems. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.03 **Duration: 40 minutes**

Binary Conversion

Learning Objectives

- Convert decimal numbers to binary
- Convert binary numbers to decimal
- Verify conversions using the doubling method

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Binary Conversion - mathematical concepts
Science: Scientific applications of Binary Conversion - scientific applications
Language: Writing about Binary Conversion builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Binary Conversion concepts through practical exercises
Advanced: Apply Binary Conversion concepts to solve complex problems independently
SEND: Practice Binary Conversion skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Binary Conversion activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Binary Conversion into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Binary Conversion. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.04 **Duration: 40 minutes**

Text Representation

Learning Objectives

- Understand ASCII encoding for English text
- Explain how Unicode supports multiple languages
- Convert text to binary using character codes

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Text Representation - mathematical concepts
Science: Scientific applications of Text Representation - scientific applications
Language: Writing about Text Representation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Text Representation concepts through practical exercises
Advanced: Apply Text Representation concepts to solve complex problems independently
SEND: Practice Text Representation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Text Representation activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Text Representation into manageable steps

Dormitory Task (Paper-and-Pencil)

Plan a 4-slide presentation about Text Representation. Write the title and 3 bullet points for each slide.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.05 **Duration: 40 minutes**

Image Representation

Learning Objectives

- Understand how images are made of pixels
- Explain color depth and resolution
- Calculate storage needed for a simple image

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Image Representation - mathematical concepts
Science: Scientific applications of Image Representation - scientific applications
Language: Writing about Image Representation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Computer with image editor
- sample images
- tool reference

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Image Representation concepts through practical exercises
Advanced: Apply Image Representation concepts to solve complex problems independently
SEND: Practice Image Representation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Image Representation activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Image Representation into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple design using only basic shapes. Label the tools you would use to create it digitally.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.06 **Duration: 40 minutes**

Audio Representation

Learning Objectives

- Understand how sound waves become digital data
- Explain sampling rate and bit depth in audio
- Calculate file size for different audio quality settings

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Audio Representation - mathematical concepts
Science: Scientific applications of Audio Representation - scientific applications
Language: Writing about Audio Representation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Computer with media software
- sample media files
- editing guide

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Audio Representation concepts through practical exercises
Advanced: Apply Audio Representation concepts to solve complex problems independently
SEND: Practice Audio Representation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Audio Representation activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Audio Representation into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a plan for a short audio or video project. List the steps and tools you would use.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.07 **Duration: 40 minutes**

Data Processing

Learning Objectives

- Understand the input-process-output cycle
- Identify different types of data processing
- Explain how computers manipulate data automatically

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Data processing transforms raw data into useful information
Science: Processing pipelines connect multiple operations
Language: Writing data processing code builds ETL skills

Digital Citizen Reminder: Apply Data Processing skills responsibly and ethically.

Resources Needed:
- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Data Processing support
On Level: Demonstrate Data Processing through practical exercises
Advanced: Apply Data Processing to solve complex problems independently
SEND: Practice Data Processing with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Processing activity
CT Skill Assessed: Decomposition - breaking down Data Processing into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Data Processing. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.08 | **Duration: 40 minutes**

Data Storage

Learning Objectives

- Identify different storage devices and their uses
- Compare primary and secondary storage
- Understand units of digital storage from bytes to gigabytes

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Storage - mathematical concepts
Science: Scientific applications of Data Storage - scientific applications
Language: Writing about Data Storage builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Data Storage concepts through practical exercises
Advanced: Apply Data Storage concepts to solve complex problems independently
SEND: Practice Data Storage skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Storage activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Data Storage into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Storage. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.09 **Duration: 40 minutes**

Data Compression

Learning Objectives

- Understand why data compression is needed
- Distinguish between lossless and lossy compression
- Identify compressed file formats in daily use

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Compression - mathematical concepts
Science: Scientific applications of Data Compression - scientific applications
Language: Writing about Data Compression builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Data Compression concepts through practical exercises
Advanced: Apply Data Compression concepts to solve complex problems independently
SEND: Practice Data Compression skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Compression activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Data Compression into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Compression. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.10 **Duration: 40 minutes**

Data Encryption

Learning Objectives

- Understand why data needs protection
- Explain the concept of encryption and decryption
- Identify everyday uses of encryption

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Data Encryption - mathematical concepts
Science: Scientific applications of Data Encryption - scientific applications
Language: Writing about Data Encryption builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 4 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Data Encryption concepts through practical exercises
Advanced: Apply Data Encryption concepts to solve complex problems independently
SEND: Practice Data Encryption skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Data Encryption activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Data Encryption into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Data Encryption. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.11 **Duration: 40 minutes**

Project - Data Analysis

Learning Objectives

- Collect and organize data from a survey
- Use a spreadsheet to enter and calculate data
- Create a chart to present findings visually

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Data Analysis - mathematical concepts
Science: Scientific applications of Project - Data Analysis - scientific applications
Language: Writing about Project - Data Analysis builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 4 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Project - Data Analysis concepts through practical exercises
Advanced: Apply Project - Data Analysis concepts to solve complex problems independently
SEND: Practice Project - Data Analysis skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Data Analysis project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Data Analysis project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 4.12 **Duration: 40 minutes**

Chapter 4 Review

Learning Objectives

- Review all concepts from Data Representation and Processing
- Demonstrate binary conversion and data representation skills
- Apply chapter knowledge to solve a real-world problem

Engage (5-7 minutes)

Teacher Activity

Play a short audio clip song or speech. Ask: 'How was this recorded? How do we share sound over distance?' Discuss how audio technology changed communication.

Student Activity

Listen to the clip. Discuss how radios, phones, and internet share audio. Share experiences with recording voice.

Explore (10-12 minutes)

Teacher Activity

Show Audacity interface or describe it. Demonstrate recording a short clip, playing it back, and adjusting volume. Explain the waveform display.

Student Activity

Observe the demonstration. Identify the record, play, stop buttons. Note how sound appears as a waveform.

Explain (10-12 minutes)

Teacher Activity

Define audio communication: sharing information through sound. File formats: MP3 compressed, WAV high quality. Microphone converts sound to electrical signals. Digital recording stores sound as numbers.

Student Activity

Take notes. Compare MP3 vs WAV: file size vs quality. Understand how microphones work.

Elaborate (8-10 minutes)

Teacher Activity

Task: 'Record a 30-second introduction about yourself. Practice speaking clearly.' If no computers, practice the script on paper and plan what to say.

Student Activity

Write a script. Practice delivery. If recording available, record and listen back. Identify one improvement.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two audio file formats. 2. What does a microphone do? 3. Why is clear speaking important for audio recording?'

Student Activity

Complete the exit ticket. Discuss answers with partner before submitting.

Cross-Curricular Connections

Mathematics: Review exercises consolidate web development concepts
Science: Chapter summaries connect HTML and CSS topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 4 Review skills responsibly and ethically.

Resources Needed:
- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 4 Review support
On Level: Demonstrate Chapter 4 Review through practical exercises
Advanced: Apply Chapter 4 Review to solve complex problems independently
SEND: Practice Chapter 4 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 4 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 4 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 4 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.01 **Duration: 40 minutes**

Introduction to Scratch

Learning Objectives

- Understand what Scratch is and its purpose
- Navigate the Scratch interface and identify key areas
- Create a simple project with a sprite and backdrop

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Block-based programming follows logical sequences - algorithm concepts
Science: Scratch coordinates use x,y plane - connecting to coordinate geometry
Language: Writing Scratch programs builds computational thinking

Digital Citizen Reminder: Share your Scratch projects responsibly and give credit to others.

Resources Needed:
- Computer with Scratch
- project worksheet
- design template

Differentiation

Approaching: Move a sprite to three positions using three move blocks
On Level: Create an animation with a sprite moving and changing costume
Advanced: Create a story with multiple sprites, backdrops, and dialogue
SEND: Click and drag blocks with teacher guidance to move a sprite

Formative Assessment

Method: Project checklist and demonstration
CT Skill Assessed: Decomposition - breaking an animation into individual sprite actions

Dormitory Task (Paper-and-Pencil)

Draw a 4-panel comic strip showing a character moving from left to right. Label what happens in each panel.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.02 **Duration: 40 minutes**

Motion Blocks

Learning Objectives

- Identify and use basic Motion blocks in Scratch
- Make a sprite move in different directions
- Control sprite movement using steps and turns

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Motion Blocks - mathematical concepts
Science: Scientific applications of Motion Blocks - scientific applications
Language: Writing about Motion Blocks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with Scratch
- block reference card
- project worksheet

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Motion Blocks concepts through practical exercises
Advanced: Apply Motion Blocks concepts to solve complex problems independently
SEND: Practice Motion Blocks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Motion Blocks activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Motion Blocks into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Motion Blocks. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.03 **Duration: 40 minutes**

Looks Blocks

Learning Objectives

- Use Looks blocks to change sprite appearance
- Switch costumes and backdrops programmatically
- Apply size and visibility effects to sprites

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Looks Blocks - mathematical concepts
Science: Scientific applications of Looks Blocks - scientific applications
Language: Writing about Looks Blocks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with Scratch
- block reference card
- project worksheet

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Looks Blocks concepts through practical exercises
Advanced: Apply Looks Blocks concepts to solve complex problems independently
SEND: Practice Looks Blocks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Looks Blocks activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Looks Blocks into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Looks Blocks. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.04 **Duration: 40 minutes**

Sound Blocks

Learning Objectives

- Add sound effects and music to Scratch projects
- Use Sound blocks to play, stop, and control volume
- Record and use custom sounds in projects

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Sound Blocks - mathematical concepts
Science: Scientific applications of Sound Blocks - scientific applications
Language: Writing about Sound Blocks builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Computer with Scratch
- block reference card
- project worksheet

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Sound Blocks concepts through practical exercises
Advanced: Apply Sound Blocks concepts to solve complex problems independently
SEND: Practice Sound Blocks skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Sound Blocks activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Sound Blocks into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Sound Blocks. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.05 **Duration: 40 minutes**

Events and Control

Learning Objectives

- Use Event blocks to start scripts
- Apply Control blocks like wait and repeat
- Understand the role of the green flag and key presses

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Event handling uses triggers - cause and effect in science
Science: Control structures (loops, conditionals) follow logical rules
Language: Writing event-driven programs builds sequential thinking

Digital Citizen Reminder: Test your programs frequently - finding errors early saves time.

Resources Needed:
- Computer with Scratch
- events reference
- practice exercises

Differentiation

Approaching: Use the green flag event to start a program
On Level: Combine events with loops for continuous animation
Advanced: Create a complex program using multiple events and conditional logic
SEND: Use the green flag and repeat block with teacher guidance

Formative Assessment

Method: Event handling demonstration
CT Skill Assessed: Algorithm Design - creating programs with events and control structures

Dormitory Task (Paper-and-Pencil)

Write the steps for a program that makes a sprite move when you click the green flag.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.06 **Duration: 40 minutes**

Variables

Learning Objectives

- Create and use variables in Scratch
- Understand variables as containers for storing values
- Use variables to track scores and counts

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Variables - mathematical concepts
Science: Scientific applications of Variables - scientific applications
Language: Writing about Variables builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Variables concepts through practical exercises
Advanced: Apply Variables concepts to solve complex problems independently
SEND: Practice Variables skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Variables activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Variables into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Variables. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.07 **Duration: 40 minutes**

Conditionals

Learning Objectives

- Understand if-then-else logic
- Use conditional blocks to make decisions in programs
- Combine conditions with sensing and operators

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: ‘Which looks more professional? Why?’ Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: ‘What if you need to calculate marks for 40 students?’ Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students’ marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: If-else statements use Boolean logic - true/false conditions
Science: Comparison operators (>, <, =) are mathematical
Language: Writing conditions builds logical expression skills

Digital Citizen Reminder: Test all possible conditions - programs should handle unexpected inputs.

Resources Needed:
- Computer with Scratch
- conditionals reference
- practice exercises

Differentiation

Approaching: Use if-else to change sprite color on key press
On Level: Create a simple game with if-else for scoring
Advanced: Build a complex game with nested conditions and multiple decision paths
SEND: Use an if statement with teacher guidance

Formative Assessment

Method: Conditional logic demonstration
CT Skill Assessed: Algorithm Design - creating programs with decision-making

Dormitory Task (Paper-and-Pencil)

Write three if-else conditions for a game. Example: If score > 10, display You Win, else display Keep Trying.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.08 **Duration: 40 minutes**

Loops

Learning Objectives

- Use repeat and forever loops
- Understand when to use count-controlled vs event-controlled loops
- Combine loops with other blocks for complex actions

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Loops - mathematical concepts
Science: Scientific applications of Loops - scientific applications
Language: Writing about Loops builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Loops concepts through practical exercises
Advanced: Apply Loops concepts to solve complex problems independently
SEND: Practice Loops skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Loops activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Loops into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Loops. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.09 | Duration: 40 minutes

Sensing

Learning Objectives

- Use Sensing blocks to detect interactions
- Detect mouse position, key presses, and sprite collisions
- Create interactive projects using sensing

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Sensing - mathematical concepts
Science: Scientific applications of Sensing - scientific applications
Language: Writing about Sensing builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Sensing concepts through practical exercises
Advanced: Apply Sensing concepts to solve complex problems independently
SEND: Practice Sensing skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Sensing activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Sensing into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Sensing. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.10	Duration: 40 minutes

Operators

Learning Objectives

- Use mathematical operators for calculations
- Apply comparison and logical operators
- Combine operators in expressions and conditions

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Operators - mathematical concepts
Science: Scientific applications of Operators - scientific applications
Language: Writing about Operators builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 5 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Operators concepts through practical exercises
Advanced: Apply Operators concepts to solve complex problems independently
SEND: Practice Operators skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Operators activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Operators into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Operators. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.11	**Duration: 40 minutes**

Project - Animation

Learning Objectives

- Apply all Scratch blocks to create an animation
- Plan and design an animation story
- Debug and refine a complete Scratch project

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Animation - mathematical concepts
Science: Scientific applications of Project - Animation - scientific applications
Language: Writing about Project - Animation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 5 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Project - Animation concepts through practical exercises
Advanced: Apply Project - Animation concepts to solve complex problems independently
SEND: Practice Project - Animation skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Animation project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Animation project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 5.12 **Duration: 40 minutes**

Chapter 5 Review

Learning Objectives

- Review all Scratch programming concepts
- Demonstrate understanding through a quiz
- Connect programming skills to real-world applications

Engage (5-7 minutes)

Teacher Activity

Show a handwritten letter and a typed letter. Ask: 'Which looks more professional? Why?' Discuss how word processors improve presentation. Show Writer interface.

Student Activity

Compare both letters. Identify differences: font, alignment, neatness. Express preference for typed version.

Explore (10-12 minutes)

Teacher Activity

Open LibreOffice Writer. Demonstrate: typing text, changing font and size, bold/italic/underline, alignment, saving a file. Students follow along.

Student Activity

Follow along on their own screens. Type a short paragraph. Apply formatting. Save the file.

Explain (10-12 minutes)

Teacher Activity

Define word processing: creating, editing, and formatting text documents. Key features: spell check, find/replace, copy/paste, tables, images. Introduce spreadsheet: 'What if you need to calculate marks for 40 students?' Show Calc interface.

Student Activity

Take notes on Writer features. Observe spreadsheet demo. Understand when to use Writer vs Calc.

Elaborate (8-10 minutes)

Teacher Activity

Task 1: Write a formal letter to the principal requesting a computer lab. Task 2: Enter 5 students' marks in a spreadsheet, use SUM to total, AVERAGE to find mean.

Student Activity

Complete both tasks. Apply formatting in Writer. Write formulas in Calc. Compare results with partner.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. Name two features of a word processor. 2. What is the difference between a word processor and a spreadsheet? 3. Write a SUM formula for cells A1 to A5.'

Student Activity

Answer all three questions. Review formulas before submitting.

Cross-Curricular Connections

Mathematics: Review exercises consolidate algorithm concepts
Science: Chapter summaries connect all algorithm topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 5 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 5 Review support
On Level: Demonstrate Chapter 5 Review through practical exercises
Advanced: Apply Chapter 5 Review to solve complex problems independently
SEND: Practice Chapter 5 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 5 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 5 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 5 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.01 **Duration: 40 minutes**

What is Software?

Learning Objectives

- Define software and distinguish it from hardware
- Identify different types of software
- Explain how software makes computers useful

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in What is Software? - mathematical concepts
Science: Scientific applications of What is Software? - scientific applications
Language: Writing about What is Software? builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of What is Software? concepts through practical exercises
Advanced: Apply What is Software? concepts to solve complex problems independently
SEND: Practice What is Software? skills with step-by-step teacher guidance

Formative Assessment

Method: Practical What is Software? activity and oral questioning
CT Skill Assessed: Decomposition - breaking down What is Software? into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about What is Software?. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.02 **Duration: 40 minutes**

System Software

Learning Objectives

- Identify system software and its types
- Understand the role of the operating system
- Explain how system software manages computer resources

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in System Software - mathematical concepts
Science: Scientific applications of System Software - scientific applications
Language: Writing about System Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of System Software concepts through practical exercises
Advanced: Apply System Software concepts to solve complex problems independently
SEND: Practice System Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical System Software activity and oral questioning
CT Skill Assessed: Decomposition - breaking down System Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about System Software. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.03 **Duration: 40 minutes**

Application Software

Learning Objectives

- Identify application software and its categories
- Understand purpose-specific software tools
- Compare application software with system software

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Application Software - mathematical concepts
Science: Scientific applications of Application Software - scientific applications
Language: Writing about Application Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Application Software concepts through practical exercises
Advanced: Apply Application Software concepts to solve complex problems independently
SEND: Practice Application Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Application Software activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Application Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Application Software. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.04 **Duration: 40 minutes**

Word Processing

Learning Objectives

- Open, create, and save documents in a word processor
- Format text with font, size, color, and alignment
- Use basic word processing features for writing

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Word Processing - mathematical concepts
Science: Scientific applications of Word Processing - scientific applications
Language: Writing about Word Processing builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Word Processing concepts through practical exercises
Advanced: Apply Word Processing concepts to solve complex problems independently
SEND: Practice Word Processing skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Word Processing activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Word Processing into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Word Processing. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.05 **Duration: 40 minutes**

Spreadsheet Basics

Learning Objectives

- Understand the structure of rows, columns, and cells
- Enter and format data in a spreadsheet
- Use basic formulas for calculations

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Spreadsheet Basics - mathematical concepts
Science: Scientific applications of Spreadsheet Basics - scientific applications
Language: Writing about Spreadsheet Basics builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Computer with spreadsheet
- practice data set
- formula reference

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Spreadsheet Basics concepts through practical exercises
Advanced: Apply Spreadsheet Basics concepts to solve complex problems independently
SEND: Practice Spreadsheet Basics skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Spreadsheet Basics activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Spreadsheet Basics into manageable steps

Dormitory Task (Paper-and-Pencil)

Draw a simple spreadsheet on graph paper with 5 columns and 6 rows. Fill in sample data.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.06 **Duration: 40 minutes**

Presentation Software

Learning Objectives

- Create slides with text, images, and backgrounds
- Apply transitions and animations to slides
- Design an effective presentation with clear structure

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Presentation Software - mathematical concepts
Science: Scientific applications of Presentation Software - scientific applications
Language: Writing about Presentation Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Computer with presentation software
- design guide
- sample slides

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Presentation Software concepts through practical exercises
Advanced: Apply Presentation Software concepts to solve complex problems independently
SEND: Practice Presentation Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Presentation Software activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Presentation Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Plan a 4-slide presentation about Presentation Software. Write the title and 3 bullet points for each slide.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.07 **Duration: 40 minutes**

Database Introduction

Learning Objectives

- Understand what a database is and its purpose
- Identify fields, records, and tables
- Recognize real-world uses of databases

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: ‘How do you think this was made?’ Explain: ‘A computer only does what you tell it — programming is giving instructions.’ Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a ‘move 10 steps’ block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: ‘Make your sprite move in a square.’ Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Database Introduction - mathematical concepts
Science: Scientific applications of Database Introduction - scientific applications
Language: Writing about Database Introduction builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Database software
- practice data set
- query reference card

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Database Introduction concepts through practical exercises
Advanced: Apply Database Introduction concepts to solve complex problems independently
SEND: Practice Database Introduction skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Database Introduction activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Database Introduction into manageable steps

Dormitory Task (Paper-and-Pencil)

Design a simple database table for storing student information. List at least 5 fields with their data types.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.08 **Duration: 40 minutes**

Utility Software

Learning Objectives

- Identify common utility software types
- Understand how utilities maintain computer health
- Use basic utilities like file manager and antivirus

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Utility Software - mathematical concepts
Science: Scientific applications of Utility Software - scientific applications
Language: Writing about Utility Software builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Utility Software concepts through practical exercises
Advanced: Apply Utility Software concepts to solve complex problems independently
SEND: Practice Utility Software skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Utility Software activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Utility Software into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Utility Software. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.09 **Duration: 40 minutes**

Software Installation

Learning Objectives

- Understand how software is installed on a computer

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Software Installation - mathematical concepts
Science: Scientific applications of Software Installation - scientific applications
Language: Writing about Software Installation builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Software Installation concepts through practical exercises
Advanced: Apply Software Installation concepts to solve complex problems independently
SEND: Practice Software Installation skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Software Installation activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Software Installation into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Software Installation. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.10 **Duration: 40 minutes**

Software Licenses

Learning Objectives

- Understand what a software license is
- Differentiate between free and paid software
- Recognize the importance of respecting software licenses

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: ‘How do you think this was made?’ Explain: ‘A computer only does what you tell it — programming is giving instructions.’ Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a ‘move 10 steps’ block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: ‘Make your sprite move in a square.’ Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Software Licenses - mathematical concepts
Science: Scientific applications of Software Licenses - scientific applications
Language: Writing about Software Licenses builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:
- Chapter 6 reference materials
- practice worksheet
- assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Software Licenses concepts through practical exercises
Advanced: Apply Software Licenses concepts to solve complex problems independently
SEND: Practice Software Licenses skills with step-by-step teacher guidance

Formative Assessment

Method: Practical Software Licenses activity and oral questioning
CT Skill Assessed: Decomposition - breaking down Software Licenses into manageable steps

Dormitory Task (Paper-and-Pencil)

Write 5 key points you learned about Software Licenses. For each point, explain why it is important.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.11 **Duration: 40 minutes**

Project - Newsletter

Learning Objectives

- Apply word processing skills to create a newsletter
- Use formatting, images, and layout effectively
- Collaborate and present the final newsletter

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Mathematical concepts in Project - Newsletter - mathematical concepts
Science: Scientific applications of Project - Newsletter - scientific applications
Language: Writing about Project - Newsletter builds technical vocabulary and communication skills

Digital Citizen Reminder: Apply computing skills skills responsibly and ethically.

Resources Needed:

- Chapter 6 reference materials
- project planning template
- self-assessment rubric

Differentiation

Approaching: Complete guided practice activities with computing skills support
On Level: Demonstrate understanding of Project - Newsletter concepts through practical exercises
Advanced: Apply Project - Newsletter concepts to solve complex problems independently
SEND: Practice Project - Newsletter skills with step-by-step teacher guidance

Formative Assessment

Method: Project rubric and self-assessment
CT Skill Assessed: Evaluation - assessing the quality and completeness of the Project - Newsletter project

Dormitory Task (Paper-and-Pencil)

Plan a Project - Newsletter project. Write the steps you would follow and what materials you need.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 6.12 **Duration: 40 minutes**

Chapter 6 Review

Learning Objectives

- Review all software concepts from Chapter 6
- Demonstrate understanding through activities
- Connect software knowledge to everyday computer use

Engage (5-7 minutes)

Teacher Activity

Play a simple Scratch animation on the projector. Ask: 'How do you think this was made?' Explain: 'A computer only does what you tell it — programming is giving instructions.' Show the Scratch interface.

Student Activity

Watch the animation. Guess how it was made. Express curiosity about creating their own programs.

Explore (10-12 minutes)

Teacher Activity

Open Scratch. Show: Stage, Sprite, Blocks palette. Demonstrate dragging a 'move 10 steps' block and clicking it. Show the sprite moving. Let students try.

Student Activity

Follow along. Drag blocks. Click to see the sprite move. Experiment with different blocks.

Explain (10-12 minutes)

Teacher Activity

Define programming: writing instructions for a computer. Scratch uses blocks that snap together like puzzle pieces. Key concepts: sequence order matters, events something happens, loops repeat.

Student Activity

Take notes. Understand that blocks must be in the right order. Identify events green flag click and loops repeat 10 times.

Elaborate (8-10 minutes)

Teacher Activity

Challenge: 'Make your sprite move in a square.' Hint: move forward, turn 90 degrees, repeat 4 times. Walk around to help stuck pairs.

Student Activity

Work in pairs. Combine move and turn blocks. Use repeat block. Test and fix until the sprite draws a square.

Evaluate (5-8 minutes)

Teacher Activity

Exit ticket: '1. What is programming? 2. What does the 'repeat' block do? 3. Why is the order of blocks important?'

Student Activity

Answer the three questions. Show their Scratch project to a neighbor.

Cross-Curricular Connections

Mathematics: Review exercises consolidate technology concepts
Science: Chapter summaries connect all emerging tech topics
Language: Writing reviews builds synthesis skills

Digital Citizen Reminder: Apply Chapter 6 Review skills responsibly and ethically.

Resources Needed:

- Computer with Python
- code reference card
- practice exercises

Differentiation

Approaching: Complete guided practice with Chapter 6 Review support
On Level: Demonstrate Chapter 6 Review through practical exercises
Advanced: Apply Chapter 6 Review to solve complex problems independently
SEND: Practice Chapter 6 Review with step-by-step teacher guidance

Formative Assessment

Method: Practical Chapter 6 Review activity
CT Skill Assessed: Decomposition - breaking down Chapter 6 Review into manageable steps

Dormitory Task (Paper-and-Pencil)

Write a simple Python program on paper that demonstrates Chapter 6 Review. Include comments.

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson:

Lesson 1.1 **Duration: 40 minutes**

What is ICT?

Learning Objectives

- Define ICT and identify its components
- Recognize ICT devices used in daily life
- Understand the importance of ICT in modern society

Regional Metaphor: Grain Sorting as Data Processing

Teacher Activity

Ask students: "How do you sort grains at home? What steps do you follow?"

Guide discussion to connect grain sorting to data processing:
- Unsorted grains = Raw data
- Sorting process = Data processing
- Sorted grains = Information

Student Activity

Share their experiences of helping at home with grain sorting.

Give examples of different types of grains they know.

Materials Needed:
- Real grains (rice, wheat) for demonstration
- Chart paper and markers
- ICT textbook

Understanding ICT

Engage (5-7 minutes)

Teacher Activity

Show pictures of ICT devices: computer, mobile phone, tablet, television.

Ask: "How are these devices similar? How are they different?"

Student Activity

Observe the pictures and discuss in pairs.

Share observations with the class.

Explore (10-12 minutes)

Teacher Activity

Explain ICT = Information and Communication Technology

Show the components of ICT:
- Information: Data, text, images, audio, video
- Communication: Sharing, transmitting, receiving
- Technology: Tools, devices, software

Student Activity

Take notes in their notebooks.

Identify ICT components in their daily life.

Explain (10-12 minutes)

Teacher Activity

Live demonstration: Show how a computer processes information.

Use the grain sorting metaphor:
1. Input: Raw grains (unsorted)
2. Processing: Sorting process
3. Output: Sorted grains
4. Storage: Container to store sorted grains

Student Activity

Follow the demonstration.

Draw a diagram showing input ⟶ processing ⟶ output ⟶ storage.

Elaborate (8-10 minutes)

Teacher Activity

Assign Parson's Problem: Arrange the following steps in correct order:
- "We get sorted grains"
- "We put grains in the machine"
- "Machine sorts the grains"
- "We store sorted grains in container"

Student Activity

Work in pairs to arrange the steps.

Present their answer to the class.

Evaluate (5-8 minutes)

Teacher Activity

Conduct exit ticket: Ask students to write:
1. Full form of ICT
2. Two examples of ICT devices
3. One way ICT is useful in daily life

Student Activity

Complete the exit ticket.

- Submit before leaving the class.

Cross-Curricular Connections

Mathematics: Numbers and place value connect to binary representation
Science: Sound waves carry telephone signals
Language: Writing clear instructions mirrors algorithm design

Digital Citizen Reminder: Never share your password with anyone except your parents or teacher.

Dormitory Task (Paper-and-Pencil)

Complete the paper-and-pencil activity related to today's topic. Practice the concept without needing a computer.

Resources Needed:
- Chalkboard
- worksheet
- examples of ICT devices (if available)

Differentiation

Approaching: Use word bank with pictures
On Level: Core activity as described
Advanced: Research an ICT device at home
SEND: Paired support, visual aids

Formative Assessment

Method: Exit ticket and self-assessment
CT Skill Assessed: Decomposition — breaking down tasks into smaller parts

Teacher Reflection (Post-Lesson)

What worked well:

What to improve:

Connection to next lesson: